

This assignment will not be graded and is only for practice.

Level 1

1) Rank of a matrix A is equal to

1 point

- ☐ the row rank of the matrix A .
- ☐ the column rank of the matrix A .
- ☐ the nullity of the matrix A .
- ☐ the number of dependent variables in the system of equations $Ax = 0$.
- ☐ the number of independent variables in the system of equations $Ax = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

the row rank of the matrix A .

the column rank of the matrix A .

the number of dependent variables in the system of equations $Ax = 0$.

2) Let nullity of the matrix $A_{3 \times 5}$ be 2. Find the rank of the matrix $A_{3 \times 5}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

3) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) If A is an $m \times n$ matrix, then which of the following statements is true?

1 point

- ☐ $\text{rank}(A) + \text{nullity}(A) = m$
- ☐ $\text{rank}(A) + \text{nullity}(A) = n$
- ☐ $\text{rank}(A) + \text{nullity}(A) = mn$
- ☐ $\text{rank}(A) + \text{nullity}(A) = \max\{m, n\}$
- ☐ $\text{rank}(A) + \text{nullity}(A) = \min\{m, n\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\text{rank}(A) + \text{nullity}(A) = n$

5) Which of the following is true for a homogeneous system of linear equations $Ax = 0$?

1 point

- ☐ It may have no solution.
- ☐ It can have infinitely many solutions.
- ☐ It can have a unique solution.
- ☐ Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

No, the answer is incorrect.

Score: 0

Accepted Answers:

It can have infinitely many solutions.

It can have a unique solution.

Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

6) Which of the following options are correct for a square matrix A of order $n \times n$, where n is any **1 point** natural number?

- ☐ If the determinant is non-zero, then the nullity of A must be 0.
- ☐ If the determinant is non-zero, then the nullity of A may be non-zero.
- ☐ If the nullity of A is non-zero, then the determinant of A must be 0.
- ☐ If the nullity of A is non-zero, then the determinant of A may be non-zero.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the determinant is non-zero, then the nullity of A must be 0.

If the nullity of A is non-zero, then the determinant of A must be 0.

7) Choose the set of correct statements.

1 point

- ☐ If nullity of a 3×3 matrix is c for some natural number c , $0 \leq c \leq 3$, then the nullity of $-A$ will also be c .
- ☐ $\text{nullity}(A + B) = \text{nullity}(A) + \text{nullity}(B)$
- ☐ Nullity of the zero matrix of order $n \times n$, is n .
- ☐ Nullity of the zero matrix of order $n \times n$, is 0.
- ☐ There exist square matrices A and B of order $n \times n$, such that nullity of both A and B is 0, but the nullity of $A + B$ is n .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If nullity of a 3×3 matrix is c for some natural number c , $0 \leq c \leq 3$, then the nullity of $-A$ will also be c .

Nullity of the zero matrix of order $n \times n$, is n .

There exist square matrices A and B of order $n \times n$, such that nullity of both A and B is 0, but the nullity

of $A + B$ is n .

8) The rank of an $n \times n$ matrix all of whose entries are equal to 1 is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

9) If A is a 3×4 matrix, then which of the following options are true?

1 point

- ☐ $\text{rank}(A)$ must be less than or equal to 3.
- ☐ $\text{nullity}(A)$ must be greater than or equal to 1.
- ☐ If A has 2 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 2 columns, then $\text{nullity}(A) = 2$.
- ☐ If A has 2 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 2 columns, then $\text{nullity}(A) = 1$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\text{rank}(A)$ must be less than or equal to 3.

$\text{nullity}(A)$ must be greater than or equal to 1.

If A has 2 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 2 columns, then $\text{nullity}(A) = 2$.

Level 2

10) Let $Ax = 0$ be a homogeneous system of linear equations which has infinitely many solutions, where A is an $m \times n$ matrix (where, $m > 1, n > 1$). Which of the following statements are possible?

1 point

- ☐ $\text{rank}(A) = m$ and $m < n$.
- ☐ $\text{rank}(A) = m$ and $m > n$.
- ☐ $\text{rank}(A) = m$ and $m = n$.
- ☐ $\text{nullity}(A) = n$.
- ☐ $\text{nullity}(A) = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\text{rank}(A) = m$ and $m < n$.

$\text{nullity}(A) = n$.

$\text{nullity}(A) = 0$.

Consider the coefficient matrix A of the following system of linear equations to answer questions 11 and 12:

$$3x_1 + 2x_2 + x_3 = 0$$

$$x_1 + x_3 = 0$$

11) Which one of the following vector spaces represents the null space of A appropriately?

1 point

- ☐ $\{(-t, t, t) \mid t \in \mathbb{R}\}$
- ☐ $\{(t_1, t_2, \frac{t_2 - t_1}{2}) \mid t_1, t_2 \in \mathbb{R}\}$
- ☐ $\{(t, -t, t) \mid t \in \mathbb{R}\}$
- ☐ $\{(t_1, t_2, \frac{t_1 + t_2}{2}) \mid t_1, t_2 \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(-t, t, t) \mid t \in \mathbb{R}\}$$

12) What will be the rank of A and nullity of A ?

1 point

- ☐ $\text{rank}(A) = 3, \text{nullity}(A) = 2$
- ☐ $\text{rank}(A) = 2, \text{nullity}(A) = 1$
- ☐ $\text{rank}(A) = 1, \text{nullity}(A) = 2$
- ☐ $\text{rank}(A) = 2, \text{nullity}(A) = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\text{rank}(A) = 2, \text{nullity}(A) = 1$$

13) Let $Ax = 0$ be a homogeneous system of linear equations which has a unique solution, where $A \in \mathbb{R}^{n \times n}$. What is the nullity of A ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

14) Suppose $x_1 = 0$ solves $Ax = 0$, where $A \in \mathbb{R}^{2 \times 4}$. What is the minimum number of elements in a linearly independent subset of null space of A that also spans the set of solutions of $Ax = 0$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

15) Consider the following sets:

$$S_1 = \{(1, 2, 3), (4, 5, 6), (8, 9, 10)\}$$

$$S_2 = \{(1, 2, 1), (2, 4, 2), (1, 2, 5)\}$$

$$S_3 = \{(1, 2, 3), (4, 5, 6), (7, 8, 10)\}$$

$$S_4 = \{(1, 2), (4, 5)\}$$

$$S_5 = \{(1, 2), (4, 5), (3, 8)\}$$

Choose the set of correct options

☐ S_1 is a basis of $\mathbb{R}^3$.☐ S_2 is a basis of $\mathbb{R}^3$.☐ S_3 is a basis of $\mathbb{R}^3$.☐ S_4 is a basis of $\mathbb{R}^2$.☐ S_5 is a basis of $\mathbb{R}^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

 S_3 is a basis of $\mathbb{R}^3$. S_4 is a basis of $\mathbb{R}^2$.

Your score is: 0/15